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Sample Solutions 02

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1 Rounding for bin packing

Let $x_i = \frac{1}{2}$ for all i. Then, for any $\epsilon > 0$, packing items with sizes $(1+\epsilon)x_i = \frac{1}{2} + \frac{\epsilon}{2}$ takes n bins, versus the $n/2$ of the original items: $\alpha = 2$. This shows that the same, simple rounding as in the FPTAS for knapsack (rounding item values to a power of $(1+\epsilon)$) cannot be used to achieve a PTAS, and a more complicated rounding approach is required.

2 Target shooting

1. Let $X = \sum X_i$. The standard Chernoff bound, selecting $(1+\varepsilon)$ as the scaling parameter, gives
$$P(|X - \mathbb{E}[X]| \geq \varepsilon \mathbb{E}[X]) \leq 2\exp(-\varepsilon^2 \mathbb{E}[X]/3) = 2\exp(-\varepsilon^2 \frac{|T|}{|S|} m/3)$$
then, selecting m as suggested gives a probability of $O(\delta)$ of multiplicative error of $(1+\varepsilon)$ or more.

2. A single sample is from T with probability $\frac{|T|}{|S|}$. Thus, as long as $|T| \leq \frac{1}{2}|S|$, the probability none of $O(\frac{|S|}{|T|})$ samples is from T is
$$\left(1 - \frac{|T|}{|S|}\right)^{O\left(\frac{|S|}{|T|}\right)} = \left(\left(1 - \frac{|T|}{|S|}\right)^{\frac{|S|}{|T|}}\right)^{O(1)} \geq 4^{-O(1)}$$
which is constant.

3. Call the algorithm $O(\log 1/\delta)$ times, and return the median result. This only fails if more than half of the returned values are less than $(1-\varepsilon)$OPT or more than half are more than $(1+\varepsilon)$OPT. But the probability of landing outside the correct range is at most $\frac{1}{3}$, and that for correct range at least $\frac{2}{3}$. Thus, probability of failure with at least $O(\log 1/\delta)$ calls is at most $2^{-O(\log 1/\delta)} = O(\delta)$, as desired.

3 Counting satisfying assignments

We can propose a target-shooting algorithm which we will show to be an FPRAS faster than the DFN-COUNT algorithm as described in the lecture notes. To begin, let F be a disjunction of m clauses C_i, where each C_i is a conjunction of up to n literals. Let $f(F)$ be the number of satisfying assignments to F. By assumption, there exists an $i \in [m]$ such that $|C_i| = 10$. It follows that $f(F) \geq 2^{n-|C_i|} = 2^{n-10}$.

For our target-shooting algorithm, will sample k assignments α_j uniformly at random and count how many of these satisfy F. To show that this target-shooting algorithm is an FPRAS, we start by defining:

$$X_j = \begin{cases} 1 & F[\alpha_j] = 1, \\ 0 & \text{otherwise} \end{cases}.$$

Note that these are independent Bernoulli random variable with probability of success as follows:

$$\Pr[X_j = 1] = \frac{f(F)}{2^n} \geq \frac{2^{n-10}}{2^n} = \frac{1}{2^{10}},$$

by the previous observation of $f(F) \geq 2^{n-10}$. Define $X = \sum_{j=1}^{k} X_j$, then using linearity of expectations we can compute $\mathbb{E}[X]$ as follows:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{j=1}^{k} X_j\right] = \sum_{j=1}^{k} \mathbb{E}[X_j] = k\frac{f(F)}{2^n} \geq \frac{k}{2^{10}}. \tag{1}$$

As $f(F)$ is the quantity we would like to estimate, we can take the value of X and multiply it by $\frac{2^n}{k}$ so that in expectation it is $f(F)$. We therefore need to show that it is also ϵ-close with probability at least $3/4$. To achieve this, let us first consider the probability that $f(F)$ is *not* ϵ-close to $\frac{2^n}{k}X$ in expectation:

$$\Pr\left[\left|\frac{2^n}{k}X - f(F)\right| \geq \epsilon f(F)\right] = \Pr\left[\left|X - k\frac{f(F)}{2^n}\right| \geq \epsilon k\frac{f(F)}{2^n}\right] = \Pr\left[|X - \mathbb{E}[X]| \geq \epsilon\mathbb{E}[X]\right].$$

As X is the sum of independent Bernoulli random variables, we can use the Chernoff bound to get an upper bound for the above probability:

$$\Pr\left[|X - \mathbb{E}[X]| \geq \epsilon\mathbb{E}[X]\right] \leq 2\exp\left(-\frac{\epsilon^2\mathbb{E}[X]}{3}\right) \leq 2\exp\left(-\frac{\epsilon^2 k}{3 \cdot 2^{10}}\right),$$

where we notably use the lower bound for $\mathbb{E}[X]$ in (1). In order for the proposed algorithm to be an FPRAS, we need to show that we can choose $k \in \text{poly}(n, 1/\epsilon)$ such that the upper bound above is less than or equal to $1/4$. For this, observe the following:

$$2\exp\left(-\frac{\epsilon^2 k}{3 \cdot 2^{10}}\right) \leq \frac{1}{4} \iff k \geq 9 \cdot 2^{10} \ln 2 \cdot \frac{1}{\epsilon^2}.$$

Thus, by setting $k = O(\epsilon^{-2})$, we get:

$$\Pr\left[|X - \mathbb{E}[X]| \geq \epsilon\mathbb{E}[X]\right] \leq \frac{1}{4} \iff \Pr\left[\left|\frac{2^n}{k}X - f(F)\right| \leq \epsilon f(F)\right] \geq \frac{3}{4}.$$

What remains to be shown is to analyze the runtime of this algorithm. Observe that we can sample an assignment α_j in $\mathcal{O}(n)$ and check whether it satisfies F in $\mathcal{O}(nm)$, which gives a complexity of $\mathcal{O}(nm)$ for each of the $k \in \text{poly}(n, 1/\epsilon)$ assignments. Overall, this gives $\mathcal{O}(nm/\epsilon^2)$, thus showing that this target-shooting algorithm is indeed an FPRAS. Moreover, comparing this with DNF-COUNT as given in the lecture notes, this algorithm is indeed faster, as desired.

4 Cut Counting

We consider graph $G = (V, E)$ with edge connectivity k. First, notice that since the graph is k-edge connected, the minimum degree of the graph is k and hence the graph has at least $\frac{nk}{2}$ edges. Next, observe that the minimum cut does not decrease after contraction. To observe this, notice that any cut in the contracted graph (assuming that it has at least two vetices) has the same number of edges crossing it as the corresponding cut, obtained by replacing the contracted vertex with the original two vertices, in the original graph. Thus, no new cuts are created and the minimum cut size remains the same.

Fix an arbitrary cut $C = (S, V \setminus S)$ of size at most αk. The probability of preserving this cut after contracting a random edge is at least:

$$\frac{|E| - \alpha k}{|E|} \geq 1 - \frac{\alpha k}{\frac{nk}{2}} = 1 - \frac{2\alpha}{n}$$

Repeat this contraction $n - r$ times, where $r = \lceil 2\alpha \rceil$. Contracting the i-th edge, preserve the cut with probability $1 - \frac{\alpha k}{(n-i)}$ (because contracting does not decrease the min-cut and the above argument goes through). Hence, after $(n - r)$ contractions, the probability of preserving the cut C is at least $\prod_{i=r+1}^{n} \left(1 - \frac{2\alpha}{i}\right)$.

With r nodes remaining, in total there are $2^{r-1}-1$ different cuts exists. Thus, the probability q that the final uniformly picked cut is C is at least:

$$q \geq \frac{1}{2^{r-1} - 1} \prod_{i=r+1}^{n} \left(1 - \frac{2\alpha}{i}\right).$$

Now, notice that if α is an integer, we have $r = 2\alpha$ and consequently

$$\begin{aligned}
\prod_{i=r+1}^{n} \left(1 - \frac{2\alpha}{i}\right) &= \prod_{i=r+1}^{n} \left(\frac{i-r}{i}\right) \\
&= \left(\prod_{j=0}^{r-1} \frac{1}{n-j}\right) \cdot \left(\prod_{j=1}^{r} j\right) \\
&\geq n^{-r} \cdot (r!) \\
&\geq n^{-r} \cdot 2^{r-1}.
\end{aligned}$$

Thus, we conclude that $q \geq n^{-r}$.

Notice that at the end of the above process only one cut remains, and we can do that for every cut of size at most αk. On the other hand, sum of the probability of preserving these cuts should be at most 1. Thus, the number of cuts of size at most αk is upper bounded by $1/q$ which is at most $n^{2\alpha}$.